PHARMACOLOGY · MCQ PRACTICE

GIT Infection and Antiviral Drugs

SGL 3
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A patient with acute gastroenteritis is most concerned about which major danger of GI infections? EASY

A. Dehydration
B. Hepatic failure
C. Renal failure
D. Respiratory failure
E. Cardiac arrhythmia

2. A child presents with moderate-to-severe vomiting followed by watery diarrhoea and fever. Which virus is the most common cause of gastrointestinal viral infections? EASY

A. Norovirus
B. Rotavirus
C. Astrovirus
D. Enteric adenovirus
E. Hepatitis A virus

3. A patient with a viral GI infection asks what antiviral drug will cure it. What does the lecture state about treatment of enteric viral infections? EASY

A. Oseltamivir is always the recommended first-line specific antiviral agent in every confirmed case
B. Interferon-alpha is given routinely regardless of the specific virus identified in the stool
C. There is no specific antiviral drug — treatment is conservative (antipyretics, antiemetics, ORS)
D. Acyclovir is always the recommended first-line specific antiviral agent in every confirmed case
E. Ribavirin is given routinely regardless of the specific virus identified in the stool

4. A patient asks which type of hepatitis virus is an RNA virus and, along with HBV, is a leading cause of chronic hepatitis, cirrhosis, and hepatocellular carcinoma. What is the answer? EASY

A. Hepatitis A virus (HAV)
B. Hepatitis E virus (HEV)
C. Hepatitis D virus (HDV) alone
D. Hepatitis C virus (HCV)
E. None — only HBV causes chronic disease

5. A patient with non-typhoid Salmonella infection is 10 years old. Which antibiotic regimen is appropriate for this age group? MEDIUM

A. Ciprofloxacin 500 mg/12h for 5-7 days, which is safe in this age group
B. Doxycycline 300 mg single dose, which is first-line in children
C. Azithromycin 500 mg/24h for 7 days, which is contraindicated below 14 years
D. Rifaximin 200 mg/8h for 3 days, which is first-line for Salmonella
E. Ceftriaxone 1g IV/24h for 5 days, which is safe in children below 14 years

6. A traveler develops traveler's diarrhoea from enterotoxigenic E. coli (ETEC). Which antibiotic, among those listed, is specifically noted as safe in children below 14 years, breastfeeding, and pregnancy? MEDIUM

A. Azithromycin 500 mg/24h for 1-3 days
B. Ciprofloxacin 500 mg/12h for 1-3 days
C. Rifaximin 200 mg/8h for 3 days
D. Doxycycline 300 mg single dose
E. Metronidazole 250 mg/6h for 7 days

7. A patient with Shigella dysentery fails first-line ciprofloxacin therapy. According to the WHO-recommended approach described, which agents form the second-line treatment? MEDIUM

A. Doxycycline 300 mg orally as a single dose in all cases
B. Ceftriaxone 1g IV/24h or azithromycin 500 mg/24h for 1-3 days
C. Rifaximin 200 mg/8h orally for 3 days in all cases
D. Oral metronidazole 250 mg/6h regardless of culture results
E. Oral vancomycin 125 mg/6h regardless of culture results

8. A patient is diagnosed with Campylobacter jejuni gastroenteritis. What is the WHO-recommended treatment? MEDIUM

A. Ciprofloxacin 500 mg/12h for 5-7 days
B. Doxycycline 300 mg single dose
C. Azithromycin 500 mg/24h for 3 days
D. Oral metronidazole 250 mg/6h
E. Rifaximin 200 mg/8h for 3 days

9. A patient is prescribed antibiotics for cholecystitis. Which of the following is listed as an appropriate antibacterial choice for this indication? MEDIUM

A. Oral metronidazole alone
B. Doxycycline alone
C. Azithromycin alone
D. Piperacillin/tazobactam
E. Oral vancomycin

10. A patient with chronic HBV infection asks about the main treatment goal. What does the lecture state is the primary goal of therapy? MEDIUM

A. To completely eradicate every trace of the virus from the body in all patients treated
B. To reverse existing cirrhosis in all treated patients regardless of disease stage
C. To eliminate the need for any future monitoring of liver function or viral markers
D. To prevent only acute, not chronic, HBV infection from ever developing
E. To improve survival and quality of life by preventing disease progression and hepatocellular carcinoma development

11. A patient on pegylated interferon-alpha for chronic HBV develops transient flulike symptoms and liver enzyme elevation. What is noted about these effects? MEDIUM

A. They subside with continued administration
B. They require immediate permanent discontinuation of therapy
C. They indicate treatment failure and require a dose increase
D. They are signs of irreversible bone marrow suppression
E. They require immediate liver transplantation

12. A patient with chronic HBV infection and psychiatric history is being considered for interferon-alpha therapy. What caution does the lecture highlight? MEDIUM

A. Interferon-alpha is absolutely contraindicated in every patient with any psychiatric history whatsoever
B. Caution is needed in patients with psychiatric disease, epilepsy, or thyroid disease due to risk of depression
C. No caution whatsoever is needed regardless of psychiatric history or other comorbidities
D. Interferon-alpha cures underlying psychiatric disease as an additional secondary benefit of therapy
E. Psychiatric history is only ever relevant for nucleoside analogues, never for interferon

13. A clinic is treating children with acute watery diarrhoea using oral rehydration therapy. Which ORS formulation is recommended by WHO and UNICEF since 2002, and what benefit does it offer over the original formula? MEDIUM

A. The original high-osmolarity ORS, which is now preferred because it rehydrates faster
B. Sodium bicarbonate-based ORS, which has fully replaced citrate-based formulas
C. Low-osmolarity ORS (reduced glucose and sodium chloride content), which reduces stool output, vomiting, and the need for IV therapy
D. Super ORS with added amino acids, which replaced all other formulations
E. Plain glucose solution without electrolytes, which avoids hypernatraemia

14. A patient with chronic HBV infection is switched to tenofovir after developing resistance to lamivudine and entecavir. What property of tenofovir supports this choice? MEDIUM

A. It is only effective against wild-type, non-resistant virus
B. It must always be combined with lamivudine to be effective
C. It is contraindicated after any prior nucleoside analogue therapy
D. It retains activity against lamivudine- and entecavir-resistant hepatitis virus isolates, with low resistance overall
E. It has no activity against resistant hepatitis virus strains

15. A patient is prescribed oral sofosbuvir for chronic HCV. What is distinctive about how this prodrug reaches its site of action? HARD

A. It must be activated by renal enzymes before reaching the liver via the bloodstream
B. It is active in its parent form and requires no intracellular phosphorylation at all
C. It is activated exclusively in the small intestine before systemic absorption
D. It requires co-administration with a protease inhibitor for activation
E. It is triphosphorylated within hepatic cells, with activation occurring during first-pass metabolism directly at the liver, its site of action

16. A patient with compensated chronic HCV who has previously discontinued interferon therapy due to adverse effects is being considered for an alternative regimen. Why might sofosbuvir-based therapy be an excellent alternative? HARD

A. It can be administered without the need for interferon therapy
B. It requires combination with high-dose interferon to be effective at all
C. It is only effective when combined with ribavirin and interferon together
D. It is contraindicated in any patient who has previously used interferon
E. It has the same contraindications as interferon therapy

17. A patient taking ribavirin for chronic HCV asks whether taking it with food affects absorption. What does the lecture state about this? HARD

A. Bioavailability is unaffected by food or antacids
B. Oral bioavailability (64%) increases with high-fat meals and decreases with co-administration of antacids
C. Bioavailability decreases with high-fat meals and increases with antacids
D. Ribavirin must be taken on an empty stomach for any absorption to occur
E. Antacids are required for ribavirin to be absorbed at all

18. A patient is started on Vosevi, a fixed-dose combination for chronic HCV without cirrhosis or with mild cirrhosis. What three components make up this combination, and what do they target? HARD

A. Sofosbuvir, ribavirin, and interferon alfa, all targeting host immune activation
B. Lamivudine, entecavir, and telbivudine, all targeting reverse transcriptase
C. Sofosbuvir and velpatasvir block viral RNA polymerase enzymes, while voxilaprevir blocks the viral protease enzyme
D. Boceprevir, telaprevir, and atazanavir, all targeting viral RNA polymerase
E. Adefovir, tenofovir, and ribavirin, all targeting the viral protease enzyme

19. A patient taking telaprevir for chronic HCV is advised to take it with food. What is the purpose of this instruction, and what adverse effects should be monitored? HARD

A. Food prevents absorption entirely; adverse effects include only mild headache
B. Food has no effect on absorption; the main risk is severe nephrotoxicity
C. Food is required to prevent a fatal arrhythmia; no other adverse effects are noted
D. Food improves absorption; adverse effects to monitor include anemia, rashes, lipodystrophy, liver toxicity, and diabetes
E. Food reduces drug levels intentionally; the only risk is transient insomnia

20. A patient taking Vosevi is also started on rifampicin for an unrelated infection. What should be anticipated? HARD

A. No interaction is expected between Vosevi and rifampicin
B. Rifampicin will significantly increase Vosevi's antiviral efficacy
C. This combination is specifically recommended to reduce resistance
D. Rifampicin only interacts with sofosbuvir-based regimens, not Vosevi
E. This combination must be avoided, as Vosevi must not be used together with rifampicin

✓ Answer key

Q	Answer	Difficulty	Why
1	**A**	EASY	Dehydration is the main danger of gastrointestinal infections, so rehydration (ORT) is the priority.
2	**B**	EASY	Rotavirus is the most common cause of gastrointestinal viral infections, characterized by moderate to severe vomiting followed by watery diarrhoea and fever.
3	**C**	EASY	There is no specific antiviral drug for enteric viral infections — treatment is conservative: antipyretics for fever, antiemetics for vomiting, and ORS for dehydration.
4	**D**	EASY	Hepatitis B (DNA virus) and hepatitis C (RNA virus) are the most common causes of chronic hepatitis, cirrhosis, and hepatocellular carcinoma.
5	**E**	MEDIUM	Ciprofloxacin is contraindicated in children below 14 years, breastfeeding, and pregnancy for non-typhoid Salmonella; ceftriaxone is the safe alternative in these groups.
6	**A**	MEDIUM	For traveler's diarrhoea (ETEC/EPEC), azithromycin is noted as safe in children below 14 years, breastfeeding, and pregnancy, while ciprofloxacin and rifaximin are also options but not specifically flagged as safe in these groups.
7	**B**	MEDIUM	For Shigella spp (WHO recommended), ciprofloxacin is first-line, with ceftriaxone or azithromycin as second-line treatment.
8	**C**	MEDIUM	Campylobacter jejuni is WHO-recommended to be treated with azithromycin 500 mg/24h for 3 days.
9	**D**	MEDIUM	Piperacillin/tazobactam, along with meropenem, imipenem/cilastatin, tigecycline, cefazolin/cefepime, and ciprofloxacin/levofloxacin, are listed for treatment of cholecystitis.
10	**E**	MEDIUM	The main goal of therapy for chronic HBV infection is to improve survival and quality of life by preventing disease progression and consequently hepatocellular carcinoma (HCC) development.
11	**A**	MEDIUM	Flulike symptoms, transient liver enzyme elevation, alopecia, rash, and diarrhoea subside with continued interferon administration; principal dose-limiting toxicities are bone marrow suppression, severe fatigue/weight loss, and autoimmune disorders.
12	**B**	MEDIUM	Interferon can cause depression, so caution is advised in patients with psychiatric disease, epilepsy, or thyroid disease.
13	**C**	MEDIUM	Since 2002, WHO/UNICEF have recommended low-osmolarity ORS (reduced glucose anhydrous from 20 to 13.5 g/L and sodium chloride from 3.5 to 2.6 g/L versus the original formula), which reduces stool output, vomiting, and the need for IV therapy.
14	**D**	MEDIUM	Tenofovir is highly effective with low resistance compared to adefovir, and retains activity against lamivudine- and entecavir-resistant hepatitis virus isolates.
15	**E**	HARD	Sofosbuvir is a prodrug that is triphosphorylated within hepatic cells; the enzymes required for activation are present in human hepatic cells, converting it to its active metabolite during first-pass metabolism directly at its desired site of action — the liver.
16	**A**	HARD	Sofosbuvir can be administered without the need for interferon therapy, making it an excellent alternative in patients contraindicated to IFN therapy or who have stopped IFN because of adverse effects.
17	**B**	HARD	Ribavirin has an oral bioavailability of 64%, which increases with high-fat meals and decreases with co-administration of antacids.
18	**C**	HARD	Vosevi is a fixed-dose combination of sofosbuvir, velpatasvir, and voxilaprevir: sofosbuvir and velpatasvir block two different viral RNA polymerase enzymes, while voxilaprevir blocks the protease enzyme — together blocking three proteins essential for HCV multiplication.
19	**D**	HARD	Telaprevir, a protease inhibitor, should be administered with food to improve absorption; adverse effects include anemia, rashes, lipodystrophy, liver toxicity, and diabetes, and it should be combined with peg-interferon alfa and ribavirin to improve response rates.
20	**E**	HARD	Vosevi must not be used together with rosuvastatin, dabigatran, contraceptives, rifampicin, carbamazepine, phenobarbital, or phenytoin.